Mathematical models for drug diffusion through different compartments

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Abstract

This paper is an attempt to develop mathematical models to better understand drug distribution in the human body via oral. Fick's principle and the law of mass action were used to develop a model based on the diffusion process. The rate constants were selected based on the drug efficacy at different interfaces. The ordinary differential equations (ODEs) describing the rate of change of concentration in compartments, such as stomach, lumen, arterial blood, tissue medium, and venous blood were solved using MATLAB. This was used to produce graphs depicting the change of drug concentration over time. Numerical parameters were used to calculate the drug concentration in the different compartments. The drug concentration falls in the first compartment and steadily increases in the other compartments, as shown in the graphs.[1]

Schematic

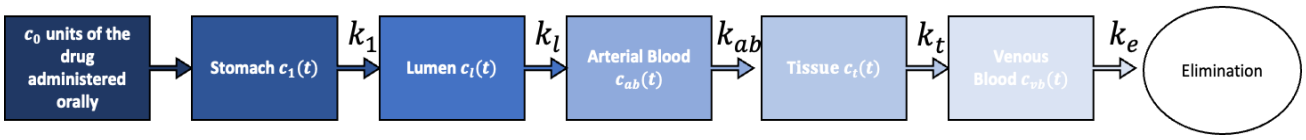

Figure 1 Drug administration through arterial blood and tissue and venous blood with initial drug intake c0.

Assumptions and Governing Equations

This principle is based on the law of conservation of mass and is known as the Balance Law.

$$\frac{dc}{dt} = input\ rate\ of\ drug - output\ rate\ of\ drug$$

(1)

We assume that the stomach takes a part of the drug, administered orally, at the rate of k_1 onto lumen, which takes a part of drug at the rate of k_l onto the arterial blood, which transports part of the drug to the tissue at the rate of k_{ab}, which then transports part of the drug to the venous blood the rate of k_t, and the finally a fraction of it gets eliminated from blood with the clearance rate of k_e. Since the blood flow in the cardiovascular system is one directional, therefore the drug administration through venous blood depicts the pattern shown in Fig. 1.

The resulting output is the concentration of the drug in the venous blood. It is evident that the drug carried out by the venous blood hits the target site through the capillary bed and the residual drug either gets eliminated or taken back by the arterial blood to the circulatory system.

The drug in venous blood is maximized as the level of drug in the venous blood increases with time and ultimately reaches to zero when the kidneys and liver excrete the drug from the body organs.[1]

Let $c_1, c_l, c_{ab}, c_t, c_{vb}$ denote the concentration of drug in the stomach, lumen, arterial blood, tissue and venous blood compartment respectively with c_0 as initial drug dosage. The mathematical formulation for the drug concentration with respect to these compartments is derived from equation **(1)** and is based on the following system of initial value problems:

$$\frac{dc_1(t)}{dt} = -k_1 c_1(t); \qquad c_1(0) = c_0 \quad \textbf{(2)}$$

$$\frac{dc_l(t)}{dt} = k_1 c_1(t) - k_l c_l(t); \quad c_l(0) = 0 \quad \textbf{(3)}$$

$$\frac{dc_{ab}(t)}{dt} = k_l c_l(t) - k_{ab} c_{ab}(t); \qquad c_{ab}(0) = 0 \quad \textbf{(4)}$$

$$\frac{dc_t(t)}{dt} = k_{ab} c_{ab}(t) - k_t c_t(t); \qquad c_t(0) = 0 \quad \textbf{(5)}$$

$$\frac{dc_{vb}(t)}{dt} = k_t c_t(t) - k_e c_{vb}(t); \qquad c_{vb}(0) = 0 \quad \textbf{(6)}$$

Parameter	**Estimation**
$k_1 = 0.9776$	Reference 1
$k_l = 0.75$	Estimated based as k_{ab} is double k_t, and k_l must be less than k_1
$k_{ab} = 0.5$	Reference 1
$k_t = 0.25$	Reference 1

$k_e = 0.005$	Reference 1
$c_1(0) = c_0$	Reference 1
$c_l(0) = 0$	Estimated based on assumption that initial concentration in the lumen = 0.
$c_{ab}(0) = 0$	Reference 1
$c_t(0) = 0$	Reference 1
$c_{vb}(0) = 0$	Reference 1

Discussion

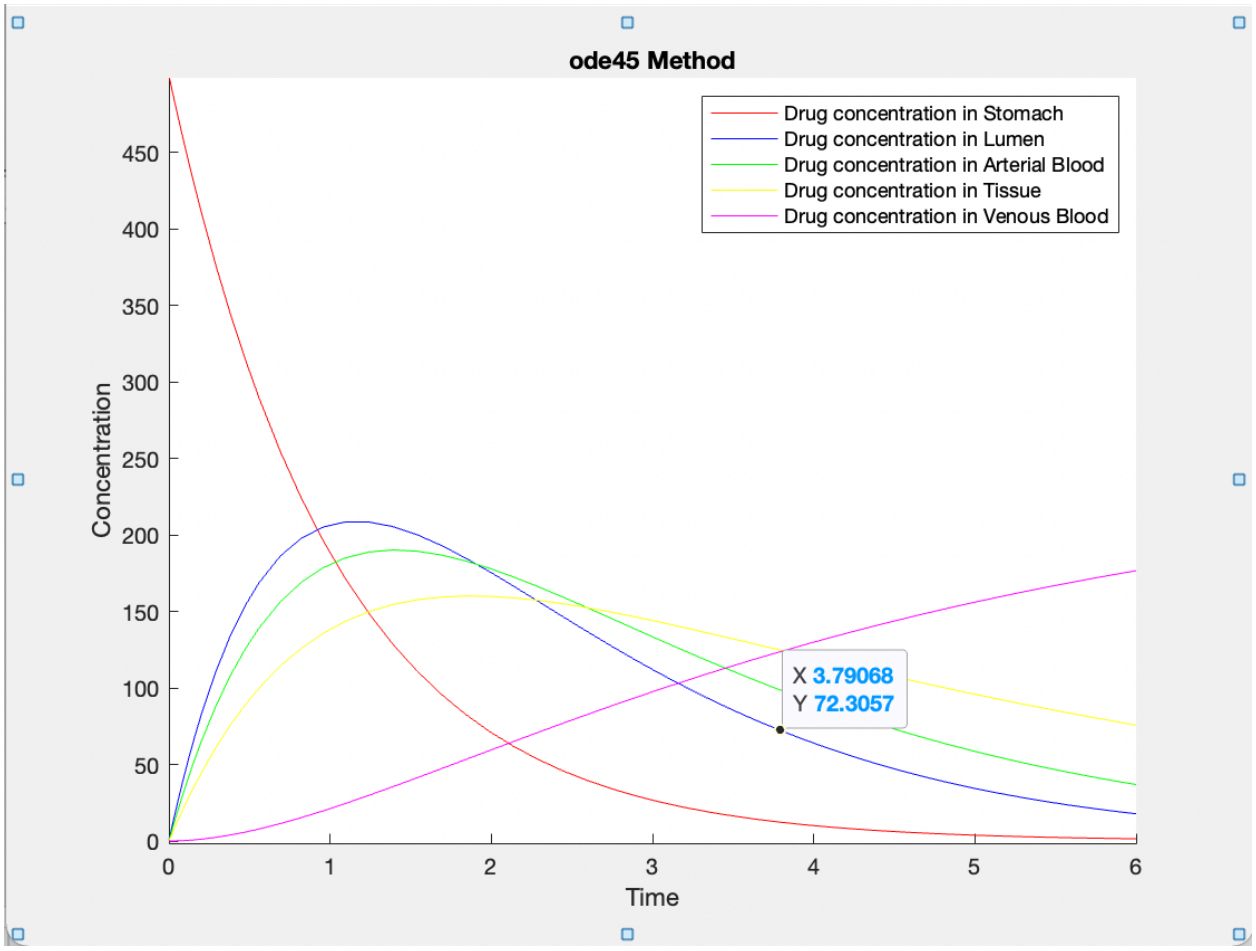

Figure 2 Drug distribution pattern in oral administration with 500 units of initial drug dosage for Figure 1. Drug concentration in the stomach decreases, as the drug is being absorbed into the lumen of the intestine, and then into the arterial blood. Drug concentration in the Arterial bloodstream decreases rapidly and while it increases in the venous blood. This is because the drug is carried out by the venous blood to the target site through the capillary bed. The Maximum drug concentration decreases for each compartment. This is due to a fraction of drug being consumed (by homogeneous reactions) in each compartment.

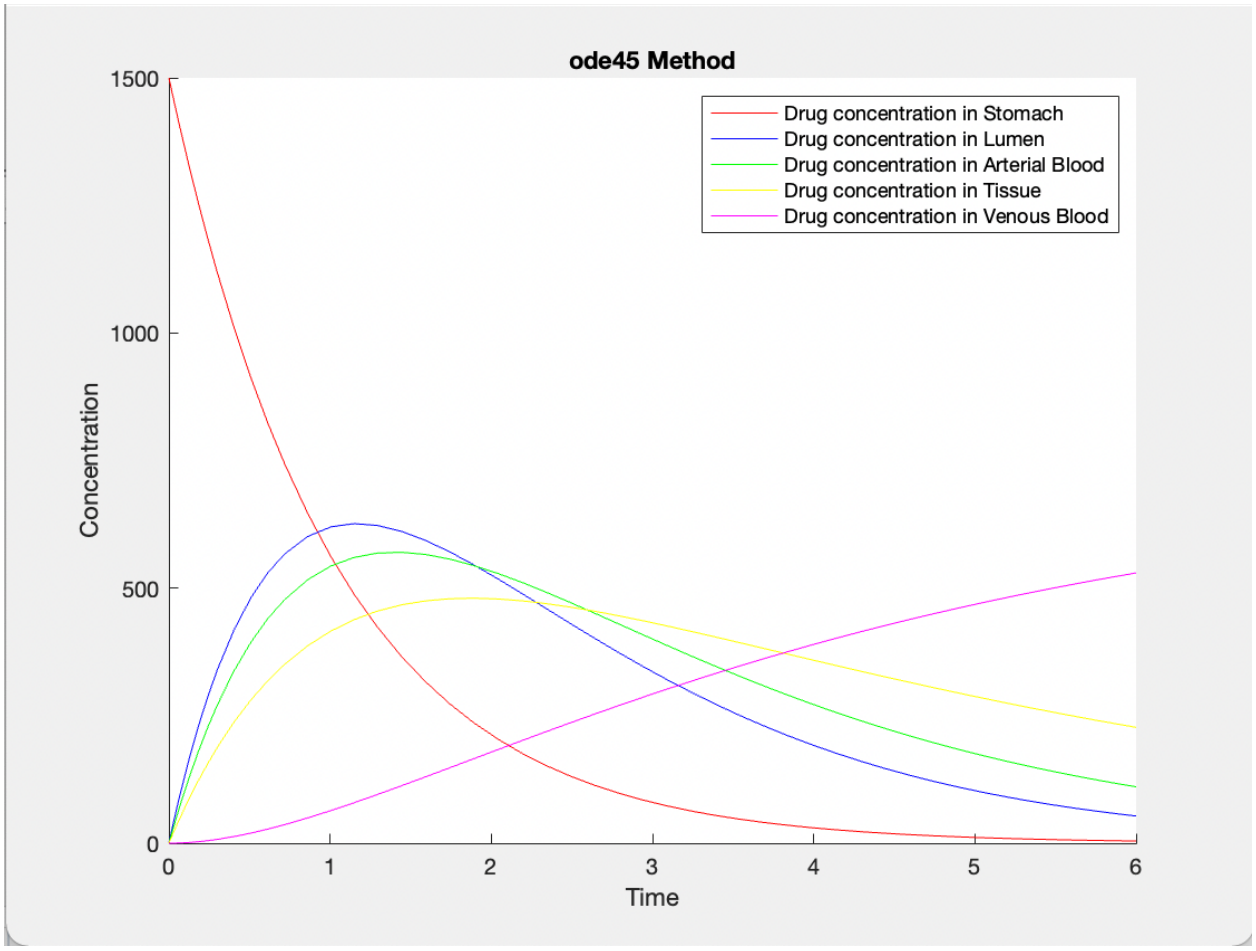

Figure 3 Drug distribution pattern in oral administration with 1500 units of initial drug dosage for Figure 1. The rate at which the drug concentration in the Arterial bloodstream decreases and the rate at which the drug in the venous blood increases stays the same as Figure 2. However, the Maximum drug concentrations for each compartment are greater than that of Figure 2. This is because the initial drug dose was tripled, and therefore even though the fraction of drug being consumed in each compartment is the same as that in Figure 2, there is more of the drug to begin with and therefore more will be remaining .

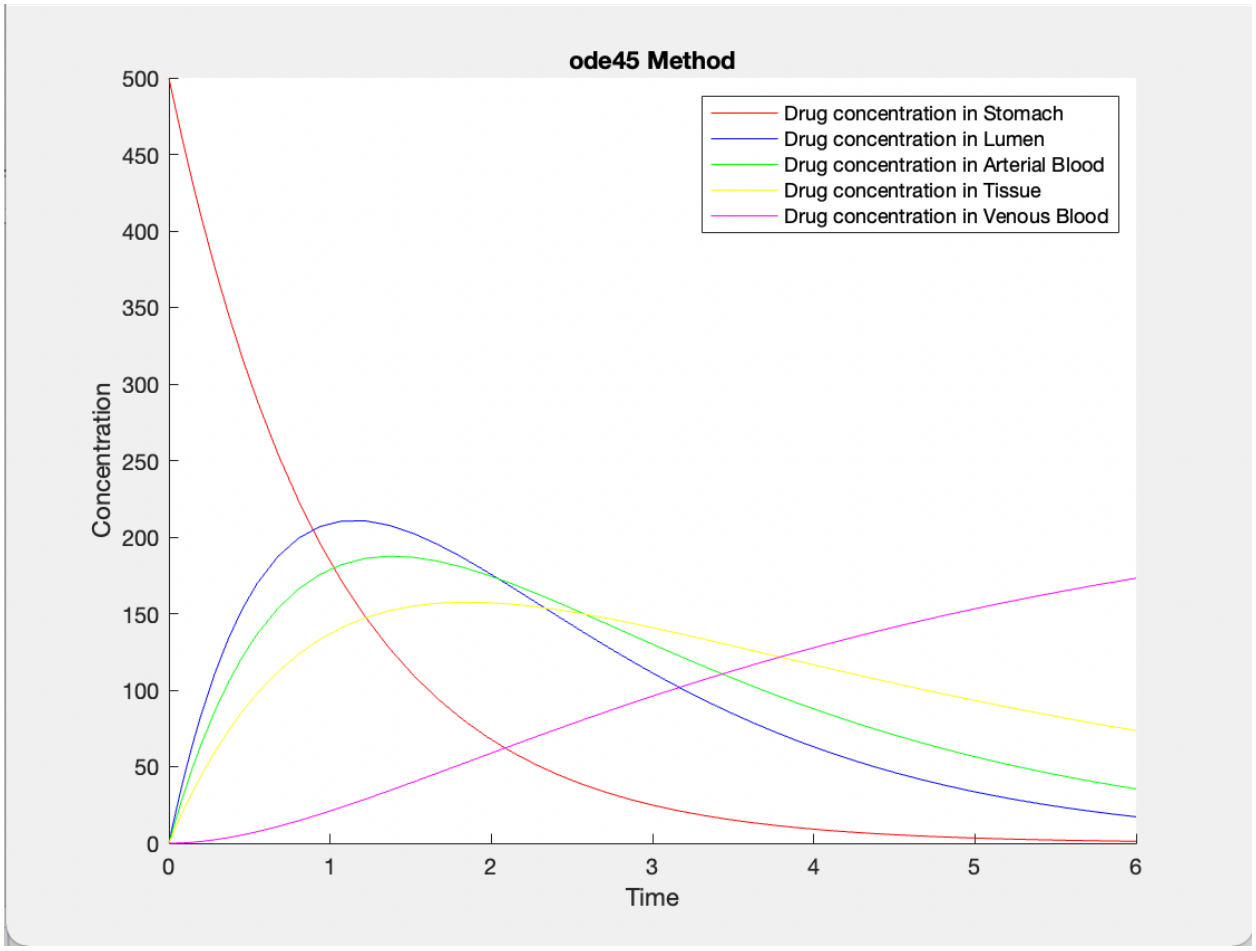

Figure 4 Drug distribution pattern in oral administration with 500 units of initial drug dosage, and ak_t= 0.5 for Figure 1. The drug concentration in the tissue medium decreases similarly while the drug concentration in the venous blood increases similarly as those in Figure 2. Varying the k_t parameter has minimal effect on the output. The rate at which the drug is being consumed by the tissue and taken up by the venous blood k_t would not have a huge impact in this model as the tissue is not the main exchanger in passive passive transcellular drug absorption.

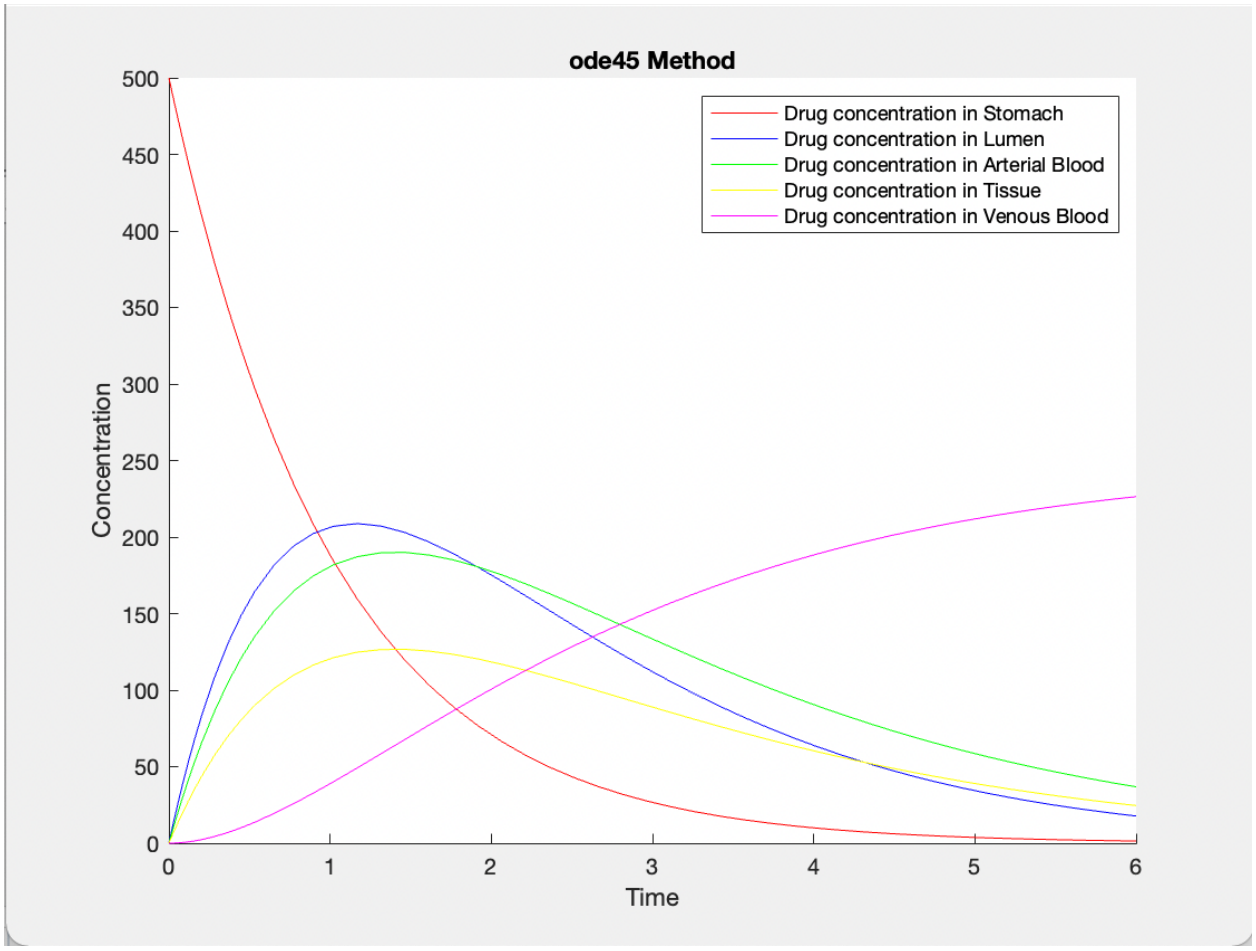

Figure 5 Drug distribution pattern in oral administration with 500 units of initial drug dosage, and ak_1= 1 for Figure 1. The drug concentration in the venous blood increases more rapidly than those in Figure 2. This is attributed to the increase in the diffusion coefficient allowing more of the drug to diffuse to the other boundary without being consumed by the homogeneous reaction.

References

1. Khanday, Mukhtar & Rafiq, Aasma & Nazir, Khalid, “Mathematical models for drug diffusion through the compartments of blood and tissue medium,” Alexandria Journal of Medicine (2016). 53. 10.1016/j.ajme.2016.03.005.